

A MAJOR PROJECT REPORT

on

A DESIGN OF 2-bit & 4-Bit SIPO & PIPO SHIFT REGISTER USING QUANTUM DOT CELLULAR AUTOMATA(QCA)

Submitted In Partial Fulfilment of the Academic Requirements of Degree of

Bachelor of Engineering In

ELECTRONICS AND COMMUNICATION ENGINEERING

BY

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CERTIFICATE

This is to certify that the Major Project Report titled, "**A NOVEL DESIGN OF 2-BIT and 4BIT SIPO & PIPO SHIFT REGISTER USING QUANTUM DOT CELLULAR AUTOMATA (QCA)**" was submitted by, **MOHD JAHANGEER BASHA (2451-21-735-023)** in partial fulfilment for the award of Bachelor of Engineering (BE) degree, in Electronics and Communication Engineering, to the Department of Electronics and Communication Engineering, MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE, an institution affiliated to OSMANIA UNIVERSITY, Hyderabad is a record of the Bonafide work carried out by them under my guidance and supervision.

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Signature of the Student

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DECLARATION

This is to certify that the Major Project Report titled, " **A DESIGN OF 2-BIT SIPO & PIPO SHIFT REGISTER USING QUANTUM DOT CELLULAR AUTOMATA (QCA)** ", being submitted by, **MOHD JAHANGEER BASHA** (2451-21-735-023) in partial fulfilment for the award of Bachelor of Engineering (BE) degree, in Electronics and Communication Engineering, to the Department of Electronics and Communication Engineering, *MATURI VENKATA SUBBA RAO (MVSr) ENGINEERING COLLEGE, affiliated to OSMANIA UNIVERSITY, HYDERABAD*, is a record of the original work carried out by the students under the project review committee of **Dr. K. Usha** (Professor Ph.D.) and **Dr. T. KAVITHA** (Associate Professor Ph.D.). Further, it is to state that the results embodied in this project report have not been submitted to any other University or Institution for the award of any Degree or Diploma.

Signature of the Students

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Abstract

Today, the implementation of computer systems and digital circuits is done using CMOS (Complementary Metal-Oxide Semiconductor) Technology. The CMOS technology has reached its physical fundamental limit due to problems such as high- power consumption, more power requirement. Due to the shortcomings in CMOS like the short channel effect, doping fluctuations, leakage current, extensive research is going on. These physical limitations of CMOS have led researchers to use the emerging technology of point cellular quantum automation (QCA).

Though various technologies have been proposed, Quantum-dot cellular automata seem to be more attractive technology. QCA is transistor less. It can replace complementary metal–oxide– semiconductor (CMOS) at nano-scale level. The most important features of this technology are very high density, low power consumption and high switching speed.

In this project, the implementation of circuits like D flip flops using QCA Method With the help of Majority and Minority Gates. The 2-Bit and 4-Bit SIPO & PIPO Shift Registers have been Designed. The simulation results have been verified using QCA Designer simulation tool. The digital technology has the advantages of reduced area, high performance. Quantum dot cellular automata require the basic components like inverters and majority gates. In this project a Designing of Flip Flops and Applications of Shift Registers has been proposed. The proposed flip flop is implemented with three-input majority gates. The Flipflop design shows much improvement in terms of cell count, occupation area and input to output delay. All Shift Registers and its Parameters are Compared i.e., Cell count, Cell area, Latency etc.

Chapter -1

Introduction

As feature sizes in CMOS continue to decrease, limitations persist, driving rapid developments in nanoscale molecular designs. QCA finds a compelling application in Static Random Access Memory (SRAM) due to its homogeneous structure, ideal for nanoscale fabrication. Noteworthy architectural innovations include parallel memories, optimizing memory cell design for efficient data movement. Quantum-Dot Cellular Automata (QCA) represents a groundbreaking technology utilizing quantum dots for digital computation. This innovative approach brings forth substantial improvements, including decreased power consumption, minimized delay, and enhanced frequency and speed in data transmission. Unlike conventional technologies such as CMOS, QCA operates without a voltage source, where electron positions dictate logical values. Moreover, QCA introduces a novel paradigm shift in clocking mechanisms, with the clock altering the barriers between dots to regulate electron tunnelling.

The distinct advantages of QCA over CMOS are noteworthy:

Achieving high operational speeds in the terahertz range. Significantly reduced power consumption (approximately 100 times less). Higher device density, approximately 10 times greater. While past studies focused on the core memory cells in CMOS devices, recent efforts emphasize the necessity for simplified clock circuitry and peripheral circuits. This paper aims to address this need by presenting a practical memory cell design for compact QCA memory solutions. Previous research highlighted shortcomings in wire crossing techniques, notably the deprecated near neighbour coplanar wire crossing. In response, this paper proposes a new SRAM design implemented in QCA, utilizing a signal distribution network (SDN) to circumvent wire crossing issues. The SDN significantly improves device performance by

relying solely on nearest neighbour interactions, enhancing thermal behaviour and fabrication tolerance. The proposed QCA-based SRAM cell demonstrates superior performance in terms of area and delay compared to CMOS-based counterparts. Notably, a 16-bit $\times$ 32-bit SRAM design achieves minimal delay and area consumption, offering efficient read/write operations and potential applications as a generalized SRAM module. This paper presents comprehensive simulations of QCA-based SRAM elements using QCA Designer software, showcasing the viability and effectiveness of the proposed design. The remainder of the paper discusses QCA fundamentals, memory cell implementations, simulation results, and concludes with future research directions. In conclusion, the integration of QCA technology in memory solutions holds immense promise for advancing VLSI technology, offering superior performance, reduced power consumption, and smaller form factors.

1.1 Basics of QCA Technology

QCA is emerging as a prominent nanotechnology, offering high switching speeds, low power consumption, and reduced area requirements. Its computational principles rely on electrostatic interactions among QCA cells. The fundamental unit of QCA computation is typically a four-dot cell. In Figure 1.1, an unoccupied dot is denoted by a white circle, while an occupied dot is represented by a black circle. Electrons within QCA cells tend to occupy diagonally opposite positions due to electrostatic interactions, creating two distinct charge distributions or logical states. These states are observable and can be leveraged to toggle between logical levels '1' and '0' for a specific bit.

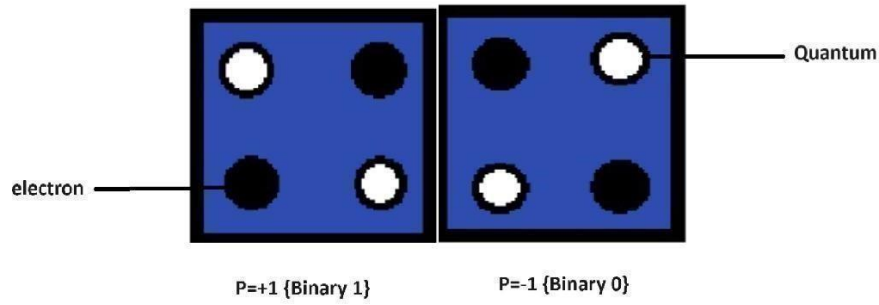


Fig. 1.1: Representation of QCA Cell

In QCA logic, we use a 3-input majority voter (MV) to process information, with its output represented as $MV(A,B,C) = A.B + B.C + A.C$ (1) .

Figure 2 shows how the majority gate functions within QCA, playing a crucial role in logical operations. Figure 3 demonstrates how a 2-bit AND gate is created using a 3-bit MV. For logical operations like OR and AND, we utilize $MV(A, B, 1)$ and $MV(A, B, 0)$ respectively. Each QCA cell consists of four quantum cells positioned at the edges of a square cell. Within these cells, two electrons can move between adjacent dots, contributing to the computation process.

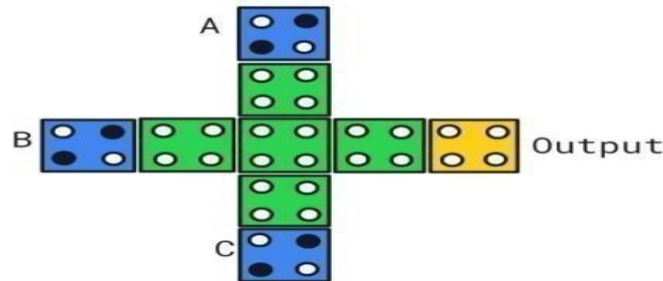


Fig.1.2: QCA Majority Gate

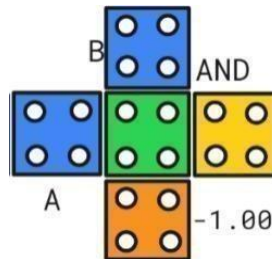


Fig. 1.3: QCA AND Gate

A QCA wire and the states of the cells in different clock zones are represented as clock-1, clock2, clock3, and clock-4 respectively. These stages are labelled as switch, hold, release, and relax as depicted in Fig. 1.4. During the switch period, the barrier potentials are low and the cells remain unpolarized. As the switch stage progresses, the QCA cells become polarized and the potential barriers increase, facilitating computation. Following the hold stage, the potential barriers remain high. During the release stage, the QCA cells depolarize as potential barriers decrease. In the relax stage, the QCA cells continue to remain depolarized as the barriers remain low.

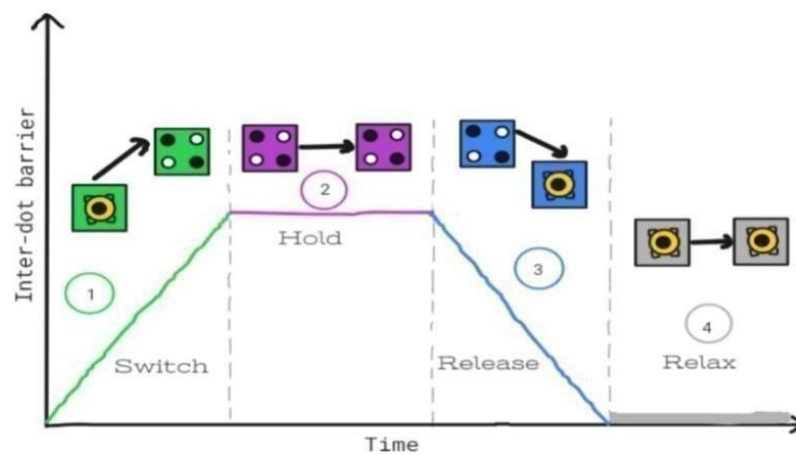


Fig. 1.4: QCA Cells Polarization of Various Clocking Zones

An established QCA inverter operates by receiving information at one end and producing an inverted output at the opposite end. The input wire splits into two separate parallel wires, and due to columbic interaction, it polarizes the cell.

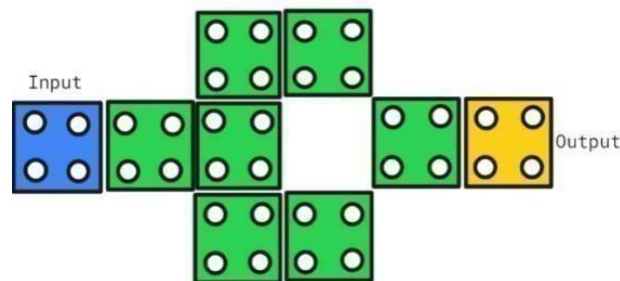


Fig-1.5 QCA NOT Gate

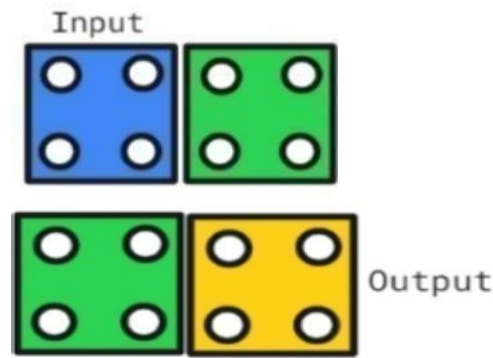
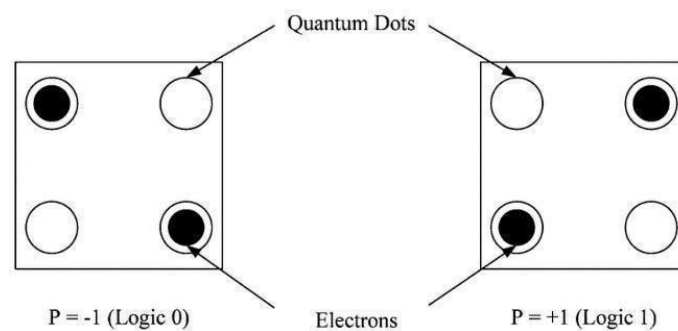


Fig.1.6 Simple QCA NOT GATE

1.2 Introduction to Quantum Cellular Automata (QCA) Technology

Quantum Cellular Automata (QCA) technology is an exciting frontier in the field of nanoelectronics, offering promising solutions to the challenges faced by traditional electronic devices. By leveraging the principles of quantum mechanics, QCA has the potential to revolutionize computing systems, enabling faster, more efficient, and more compact devices. This introduction provides a detailed overview of QCA technology, exploring its principles, applications, and current advancements.



The main aim of computer developers and industrialists is to yield smaller, swift and secure computers. So, in 1965, Gordon Moore prophesied the feat of the manufacturing industry and famed that in the middle of 1959 and 1965, the number of transistors on a die bred proportionally with time. This drift has constantly increased with the number of transistors on a die approximately doubling up every 18-24 months. This achievement has been attained mainly by

the reduction of the transistor size, abetted by the cumulative size of the die. In reality, studies specify that the physical perimeters of transistor sizing may be attained. Therefore, it may not be feasible to endure the rules of doubling the number of components in a computer chip every two years and doubling the clock rate every three years. Thus, in order to sustain trends of growing microprocessor recital, additional technologies should be premeditated.

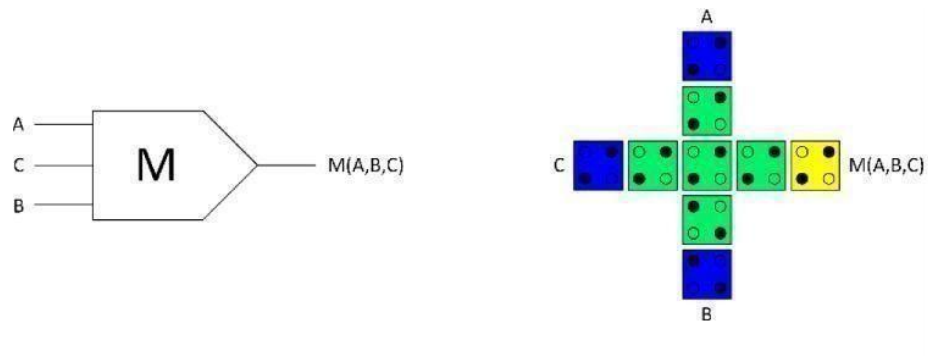


Fig.1.7. majority gate

An additional solution to the issue of device shrivelling where the performance is tainted because of growing quantum effect and extant power decadence is nanotechnology. There are many promising competitors which are being contemplated as a probable substitution to CMOS for instance Carbon Nanotubes based Transistors, Silicon Nano-wires, Quantum Dot Cellular Automata, Resonant Tunnelling devices, Spin Wave Transistors, Superconducting Electronics, amongst others. Out of these, one of the favourable solutions to performance degradation is Quantum Dot Cellular Automata (QCA). It has been projected as one of the upcoming nanotechnologies in Semiconductor Industries Association's International Roadmap for Semiconductors (ITRS).

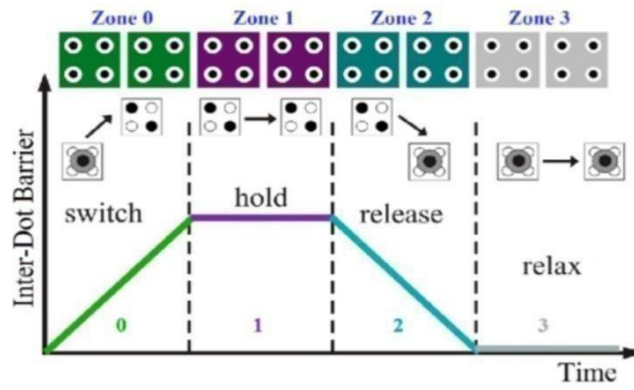


Fig .1.8. QCA clocking zones

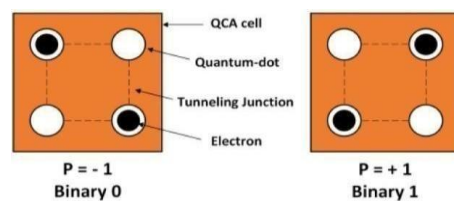
Quantum-dot Cellular Automata is an indication of demonstrating binary information on cells, where no current flows, but the collection of tiny dots where the logic state is and attaining device performance by the combination of the same cells. QCA is an imminent nano-electronic processing technology that gives the prospect to secede from FET based technology and achieve the quantum effects by means of small size. In this technology, the basic logic is no longer current but the array of small quantum dots within the QCA cell and the logic state is determined by the position of electrons via the columbic interaction. In QCA, the cells that are used for framing logic gates can be employed to construct wires conveying logic signals. The possible device density in QCA is about 900 times more than the conventional CMOS technology and also the operating frequency range is THz.

1.3 Principles of Quantum Cellular Automata (QCA)

Quantum Cellular Automata (QCA) is a revolutionary paradigm in nanoelectronics that exploits quantum mechanical principles to perform computational tasks. At its core, QCA leverages the properties of quantum bits (qubits) to encode and process information, offering potential advantages such as high-speed operation, low power consumption, and scalability beyond the limits of classical electronics. This comprehensive exploration delves into the fundamental

principles underlying QCA, including qubit representation, Coulombic interactions, logic operations, and device architectures.

1. **Quantum Bits (Qubits) in QCA:** The foundation of QCA lies in the concept of quantum bits, or qubits. Unlike classical bits, which can exist in one of two states (0 or 1), qubits can exist in multiple states simultaneously due to the phenomenon in QCA, qubits are typically represented by quantum systems such as electron spins or the presence/absence of excess electrons in quantum dots. These quantum systems can be in superposition. This property allows qubits to encode and process information in parallel, enabling potentially exponential increases in computational speed. In QCA, qubits are typically represented by quantum systems such as electron spins or the presence/absence of excess electrons in quantum dots. These quantum systems can be manipulated and controlled to perform logical operations, making them ideal candidates for information processing tasks.



2. **Coulombic Interactions:** Central to the operation of QCA devices are Coulombic interactions between qubits. Coulombic interactions arise from the electrostatic repulsion between charged particles, such as electrons in quantum dots. By modulating the strength of these interactions, QCA devices can perform logical operations and transmit signals between qubits. In a typical QCA device, qubits are arranged in a regular grid pattern, forming a cellular automaton. When qubits are in close proximity to each other, their Coulombic interactions cause them to align in a specific configuration, known as a quantum state. Controlling the arrangement of qubits and the strength of Coulombic interactions, QCA devices can perform computations by propagating signals through the grid.

- 3. Logic Operations in QCA:** QCA devices perform logical operations by manipulating the quantum states of qubits through Coulombic interactions. The most basic logical operations in QCA are AND, OR, and NOT gates, which are fundamental building blocks for constructing more complex circuits. In an AND gate, for example, the output qubit is in the ON state (1) only when both input qubits are also in the ON state. This is achieved by arranging the input qubits such that their Coulombic interactions cause them to align in a specific configuration when both are in the ON state. Similarly, OR and NOT gates can be implemented by controlling the arrangement of qubits and the strength of Coulombic interactions.
- 4. QCA Device Architectures:** QCA devices are typically constructed using a regular grid pattern of qubits, forming a cellular automaton. The basic building blocks of QCA devices include QCA cells, wires, and logic gates. QCA cells consist of multiple qubits arranged in a specific configuration to perform logical operations. These cells can be interconnected by wires, which serve as communication channels between QCA cells. Logic gates are constructed by selectively applying voltages to certain regions of the device, which modulates the strength of Coulombic interactions between qubits and allows for the manipulation of quantum states.
- 5. QCA Fabrication Techniques:** Fabricating QCA devices involves several key steps, including the deposition of quantum dots, patterning of wires, and assembly of logic gates. Quantum dots, which represent qubits in QCA devices, can be fabricated using techniques such as molecular beam epitaxy (MBE) or chemical vapor deposition (CVD), which allow for precise control over their size and position. Once the quantum dots are in place, wires are patterned onto the substrate to connect them and form the grid structure of the device. These wires serve as the communication channels between QCA cells, allowing signals to propagate through the device.

Finally, logic gates are constructed by selectively applying voltages to certain regions of the device, which modulates the strength of Coulombic interactions between qubits and allows for the manipulation of quantum states.

6. Advantages and Challenges of QCA: QCA technology offers several potential advantages over classical electronics, including high-speed operation, low power consumption, and scalability beyond the limits of classical devices. However, there are also several challenges associated with QCA, including the need for precise control over qubit states and Coulombic interactions, as well as the fabrication of QCA devices with high yields and reliability.

Despite these challenges, QCA holds great promise for revolutionizing computing and information processing, offering potential advantages such as faster operation, lower power consumption, and higher device density. As research in this field continues to progress, we can expect to see further advancements and innovations that will unlock the full potential of QCA technology and pave the way for a new era of computing and communication.

1.4 Pre-QCA Technology Era

- Before delving into the pre-Quantum-dot Cellular Automata (pre-QCA) technology era, it's important to understand what QCA technology entails. Quantum-dot Cellular Automata
- nanotechnology concept aimed at developing electronic devices at the nanoscale. Unlike traditional transistor-based technologies, QCA uses the position of electrons to represent binary information,
- potentially allowing for much denser, faster, and less power-consuming circuits.

The pre-QCA technology era essentially covers the entire history of electronic computing up until the conceptualization and early development of QCA, which gained significant attention in the late 20th and early 21st centuries. This era can be broadly categorized into several key phases, focusing on the evolution of electronic computing technologies:

1. Mechanical and Electromechanical Computing (Pre-1940s): -Before electronic computing, mechanical and electromechanical devices were the forefront of computing technology. Machines like the Babbage Difference Engine and the Analytical Engine in the 19th century, followed by electromechanical computers like the Harvard Mark I in the 1940s, laid the foundational ideas of computing.

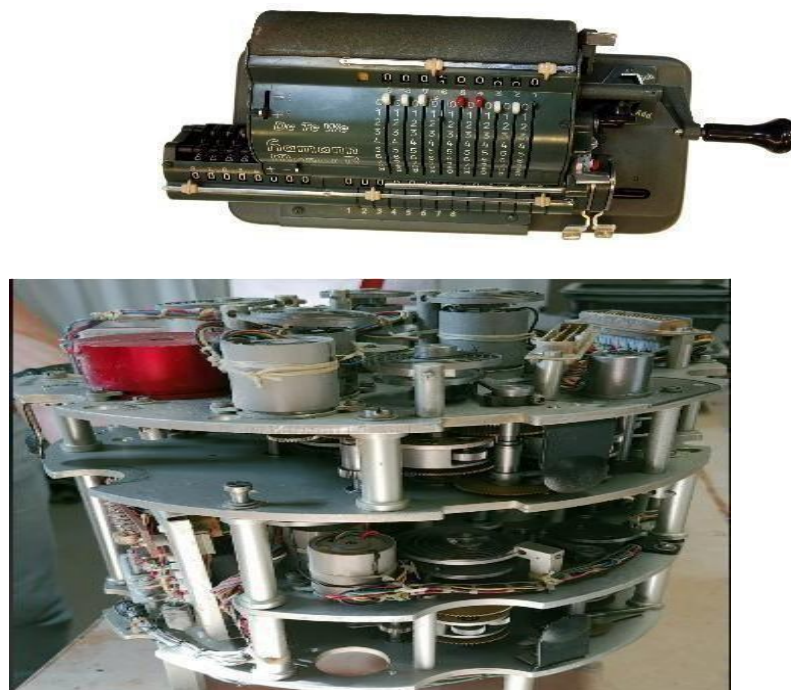


Fig.1.9. Mechanical and Electromechanical Computing

2. Vacuum Tubes Era (1940s - Late 1950s):-The invention of the vacuum tube enabled the creation of the first fully electronic computers. These machines, such as the ENIAC, were monumental in size and power consumption but revolutionary in speed and capability compared

to their mechanical predecessors. However, vacuum tubes were fragile, generated a lot of heat, and were inefficient by today's standards.



Fig.1.10 .Vacuum Tubes

3. Transistor Era (Late 1950s - 1970s):-The invention of the transistor in 1947 (with widespread adoption in the following decades) dramatically changed the landscape of electronics. Transistors were smaller, more reliable, and more efficient than vacuum tubes. This era saw the development of the first integrated circuits (ICs) and laid the groundwork for the microelectronics revolution.

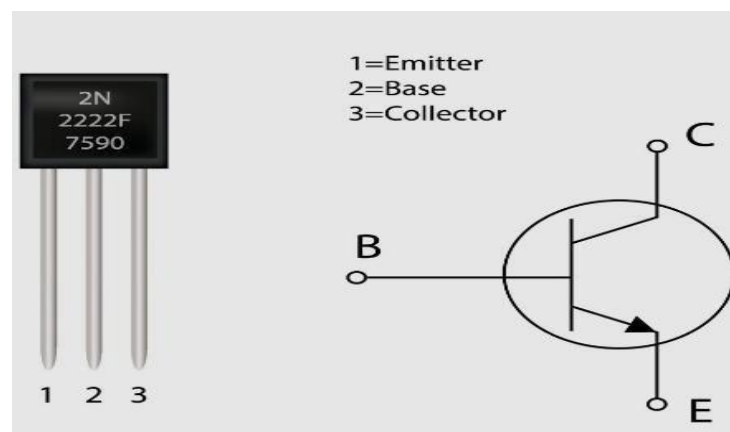


Fig.1.11 .Transistor

4. Microelectronics and the Microprocessor (1970s - Late 1990s):- By the late 1990s, 32-bit microprocessors dominated the consumer market, with clock speeds increasing by more than tenfold between 1990 and 1999.¹ This period also saw the emergence of 64-bit processors, setting the stage for the next era of computing advancements. Exponential increases in computing power and efficiency.

5. Early Nanotechnology and Post-Silicon Research (Late 1990s -

Present: -As the limits of Moore's Law became more apparent and the physical limitations of silicon-based transistors started to be reached, research into alternative computing paradigms, including QCA, gained momentum. This period is characterized by explorations into nanotechnology, quantum computing, and other post-silicon technologies aimed at overcoming the limitations of traditional transistor-based electronics.

The transition to the QCA and similar technologies represents an ongoing evolution in computing, aiming to address the physical and practical limitations of current technologies by exploring fundamentally different principles of operation. While QCA is still in the research and development phase and not yet widely adopted, it embodies the continuous quest for innovation that has characterized the entire history of computing technology.

1.6 Post-QCA Technology Era

In the realm of computing and electronic devices, the quest for more efficient, faster, and smaller components is relentless. Quantum-dot Cellular Automata (QCA) emerged as a revolutionary technology, proposing a significant departure from traditional transistor-based approaches. By

employing the principles of quantum mechanics, QCA enabled the development of devices with potentially higher speed and lower power consumption. However, as we venture into the postQCA era, new technologies are emerging, promising to further push the boundaries of computing efficiency, speed, and miniaturization. This project explores these technologies, their principles, and their potential impacts on the future of computing and electronics.

Spintronics: -One of the promising post-QCA technologies is spintronics (spin electronics), which exploits the spin of electrons in addition to their charge for information processing. Unlike QCA, which relies on the positioning of electrons, spintronics offers a way to store and manipulate information that could be more robust against temperature fluctuations and easier to integrate with existing semiconductor technologies. Spintronic devices promise not only to enhance computing power and reduce energy consumption but also to enable new forms of quantum computing and storage devices.

Photonic Computing: - Moving beyond electrons, photonic computing represents another frontier in the post-QCA era. This approach uses photons, the basic units of light, for computing operations. Photonic computing can potentially surpass the speed limitations of electronic devices, as photons travel faster than electrons and do not generate heat. This technology could lead to ultra-fast, energy-efficient computers and significantly impact data transmission technologies, making it a strong contender in the post-QCA landscape.

Topological Insulators: - Topological insulators are materials that act as insulators in their interior but conduct electricity on their surface. This unique property, stemming from quantum

mechanics, opens up new possibilities in computing, especially in the development of more efficient transistors and memory devices. In the post-QCA era, topological insulators could pave the way for devices that are less prone to overheating and more energy-efficient, addressing some of the critical limitations of both traditional and QCA based technologies.

Molecular Electronics: - Lastly, molecular electronics explores the use of individual molecules or nanoscale assemblies of molecules as electronic components. This approach could lead to the ultimate miniaturization of electronic devices, far beyond what QCA and traditional semiconductor technologies can achieve. Molecular electronics holds the potential for creating devices with unprecedented densities of components and pathways, offering a significant leap in computing power and efficiency. The post-QCA era is marked by a diverse and promising range of technologies, each with the potential to overcome the limitations of current computing paradigms. From spintronics and photonic computing to topological insulators and molecular electronics, these emerging technologies offer exciting prospects for the future of computing and electronic devices. As research and development continue, it is likely that a combination of these technologies will shape the next generation of computing, leading to devices that are faster, more efficient, and more capable than ever before. For engineers and technologists, understanding these emerging technologies is crucial for driving innovation and addressing the ever-growing demands for computing power and efficiency in a range of applications.

1.5 The Basics of Majority Gate

The fundamental structure of majority gate contains minimum five QCA cells to represent it [3] [12]. Four of them are in four different directions making themselves at the angle of 90°

with the nearest neighboring cells on the substrate and there is one cell which is in the middle of them. The Boolean expression which is suitable to represent the majority gate is – $M(A, B, P) = A.B. + A.P + B.P$

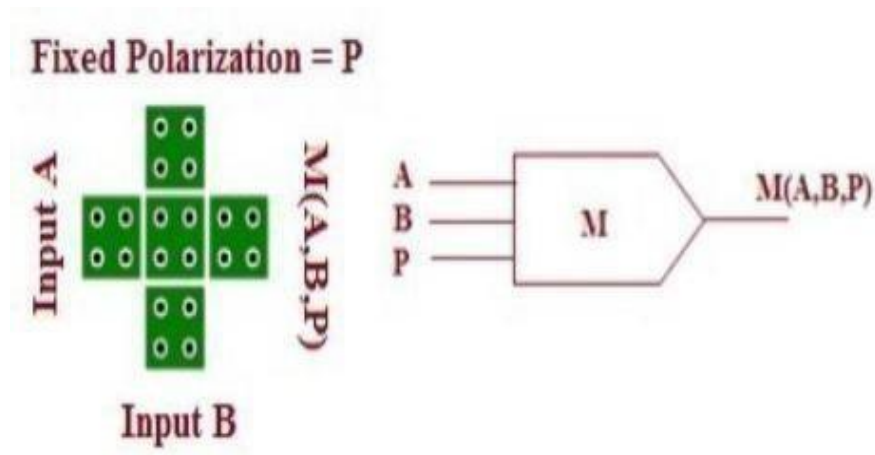


Fig.1.12: Majority Gate in QCA Where the first two terms A and B represent the inputs from the majority gate and P is the fixed polarization.

Chapter 2

Literature Survey

A flip-flop is a circuit that has two stable states and can be used to store state information. A flip flop is a binary storage device. It can store binary bit either 0 or 1. It has two stable states HIGH and LOW i.e. 1 and 0. A flip-flop is usually controlled by control signals that can include a clock signal. The outputs usually include the complement as well as the normal output. Flipflops can be either simple (transparent or opaque) or clocked (synchronous or edge-triggered). Although the term flipflop has historically referred generically to both simple and clocked circuits, in modern usage it is common to reserve the term flip-flop exclusively for discussing clocked circuits; the simple ones are commonly called latches. The basic function of flip flop is to store data. They can be used to keep a record of what value of variable (input, output or intermediate). Flip flop are also used to exercise control over the functionality of a digital circuit i.e. change the operation of a circuit depending on the state of one or more flip flops. Flipflops are sequential circuits whose output depends both on the present input as well as the past output[12].It is a one bit binary storage device capable of storing binary information '1' or '0'.In this paper, most commonly used three types flip-flops namely D,T & JK flipflop are designed from the derived equations of SR flip-flop.

History

The first electronic flip-flop was invented in 1918 by William Eccles and F. W. Jordan. It was initially IJIRT 101108 called the Eccles–Jordan trigger circuit and consisted of two active elements (vacuum tubes). Such circuits and their transistorized versions were common in computers even after the introduction of integrated circuits, though flip-flops made from logic

gates are also common now. Early flip-flops were known variously as trigger circuits or multivibrators. According to P. L. Lindley, a JPL engineer, the flip flop types discussed below (RS, D, T, JK) were first discussed in a 1954 UCLA course on computer design by Montgomery Phister, and then appeared in his book Logical Design of Digital Computers.

Implementation

Flip-flops can be either simple (transparent or asynchronous) or clocked (synchronous); the transparent ones are commonly called latches. The word latch is mainly used for storage elements, while clocked devices are described as flip-flops. Simple flip-flops can be built around a pair of cross-coupled inverting elements: vacuum tubes, bipolar transistors, field effect transistors, inverters, and inverting logic gates have all been used in practical circuits. Clocked devices are specially designed for synchronous systems; such devices ignore their inputs except at the transition of a dedicated clock signal (known as clocking, pulsing). Clocking causes the flip-flop to either change or retain its output signal based upon the values of the input signals at the transition. Some flip-flops change output on the rising edge of the clock, others on the falling edge. Since the elementary amplifying stages are inverting, two stages can be connected in succession (as a cascade) to form the needed noninverting amplifier.

Proposed Expression to Design All Flip-flops

2.1 D Flip-Flop

The D flip-flop is called Delay (D) flip-flop as it introduces a delay between input and output.

We have easily constructed D flip-flop from our proposed logic just by replacing A by D and B

When the pulse of the clocking is provided to the flip-flop circuit, the output signal starts toggle that is half the frequency of the input T. Construction of frequency dividers, binary counters and for simple binary addition is useful. The proposed T flip-flop circuit is a simplified method of the all the previous designs. The proposed majority gate structure of the T flip-flop is shown in the Fig.

2.4. The characteristic table of the flip-flop circuit is shown in the table.2.4. The proposed T flipflop layout schematic has one T input, one clock input, four negative fixed polarizations, and one positive fixed polarization.

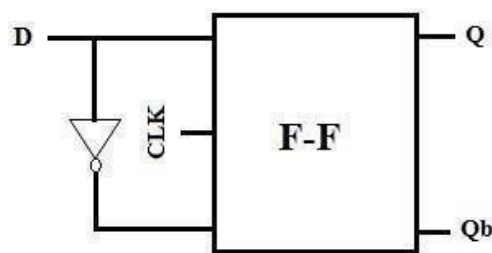


Fig 2.4: block diagram of D flip-flop majority gate structure

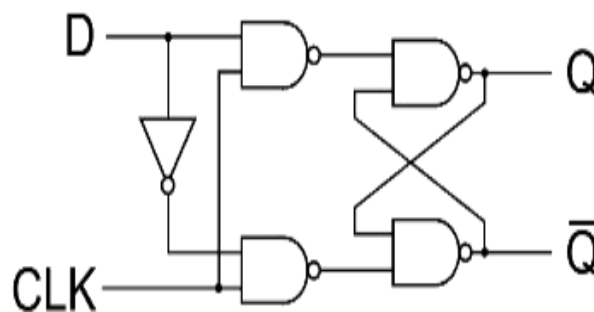


Fig.2.5: block diagram of the D flip-flop circuit

Q	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

Table.2.4 Characteristic Table of D flip-flop

2.2 Application of flip flop

Data Storage: A flip flop store one bit at a time in digital circuit. In order to store more than one bit flip flop can be connected in series and parallel called registers. A register is simply a data storage device for a number of bits in which each flip flop store one bit of information (0 or 1). Thus a 4 bit register consists of 4 individual flip flops, each able to store one bit of information at a time.

Data Transfer: Flip flops can also be used extensively to transfer the data. For this purpose shift register is used. A shift register is a register which is able to shift or transfer its content within itself without changing the order of the bits. It may be designed to shift or transfer data either left or right. The data is shifted or transferred one bit at a time, when a clock pulse is applied. The shift register can be used for temporary storage of data. The shift register is used for multiplication and division where bit shifting is required. The shift register can be built using RS, JK or D flip flops.

Counter: Another major application of flip flops is a digital counter. It is used to count pulses or events and it can be made by connecting a series of flip flops. Counter can count up to 2^n . Where n is the number of flip flops.

Frequency Division: Flip flops can divide the frequency of periodic waveform. When a pulse wave is used to toggle a flip flop, the output frequency becomes one half the input frequency. The output of each flip flop is half the frequency of an input. If the input frequency is 160 KHz then output of each flip flop would be so after first flip flop, 80 after second flip flop and 40 after third flip flop. Input frequency 160 KHz Frequency of first flip flop 80 KHz Frequency of 2nd flip flop 40 KHz Frequency of 3rd flip flop 20 KHz .

2.3 Complexity Of The Proposed Flip-flops

TABLE 2.6: CALCULATIONS OF CELL COUNTAUF, O-COST & LATENCY OF THE DESIGNED FLIP-FLOPS

QCA Structure	Length Covered (nm)	Width Covered (nm)	Cell Count (cells)	Operation Cost(O-Cost)	Net Area, (nm ²)	Total Area, (L×W) (nm ²)	Area Utilization Factor(AUF = Net Area/ Total Area)	Latency
SR flip-flop	258	158	38	38	12,312	40,764	0.302	1.50
D flip- flop	238	178	43	43	13,932	42,364	0.328	1.25
T flip- flop	258	258	81	81	26,244	66,564	0.394	1.50
JK flip-flop	278	258	78	78	25,272	71,724	0.352	1.50

2.4 CALCULATIONS OF KINK ENERGY

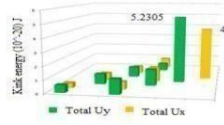
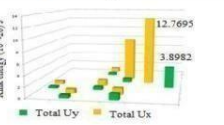
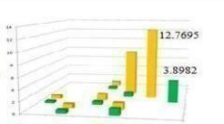
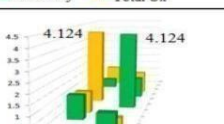
Flip-flops	Kink energy of Output cell		3-D plot of Total Kink energy (Total kink energy for electron $x = U_x$, Total kink energy for electron $x = U_y$)	
SR flip-flop	For Output = '1'	For Output = '0'		
D Flip-flop	For Output = '1'	For Output = '0'		
T Flip-flop	For Output = '1'	For Output = '0'		
JK Flip-flop	For Output = '1'	For Output = '0'		

TABLE 2.7: CALCULATIONS OF KINK ENERGY OF THE OUTPUT

CELLS OF SR, D, T, &JK FLIP-FLOPS RESPECTIVELY

2.5 ENERGY DISSIPATION

The energy dissipation [7, 11] due to change in polarization can easily be computed from the equation.

$$\leq \left[\frac{2\gamma_{new}}{E_k} \left(\frac{p_0}{\bar{p}_{old}} \gamma_{old} - \frac{p_n}{\bar{p}_{new}} \gamma_{new} \right) + \frac{E_k \bar{p}_{new}}{2} (p_0 - p_n) \right] \dots (3)$$

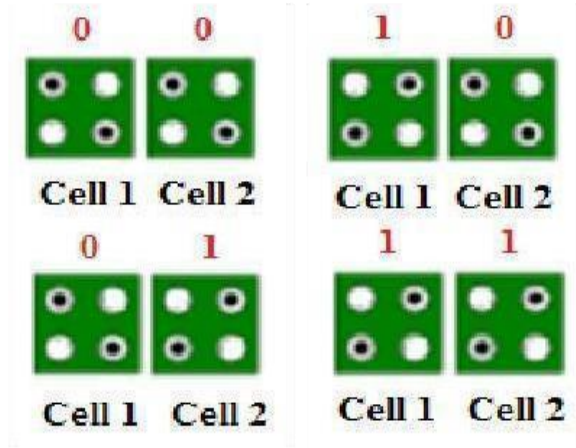


Figure 2.6 : Energy Dissipation of 0→0, 0→1, 1→0 & 1→1 transitions

Figure 2.6 shows four different combinations of polarizations from 0→0, 0→1, 1→0 & 1→1. From the four different transition of polarization we have found out corresponding the energy dissipations. Energy dissipation of 0→0 & 1→1 transition will be same but not same for the transition of 0→1 & 1→0. Similarly, energy dissipation of 0→1 & 1→0 transition will be same but not same for the transition of 0→0 & 1→1. If we plot the energy dissipation graph (vs γ), then we have found out a increasing graph of energy dissipation which is parallel to each

other's. Here, we considered that the clock energy during switching event & are same and then we plot varying γ from $0.5E_k$ to $2E_k$.

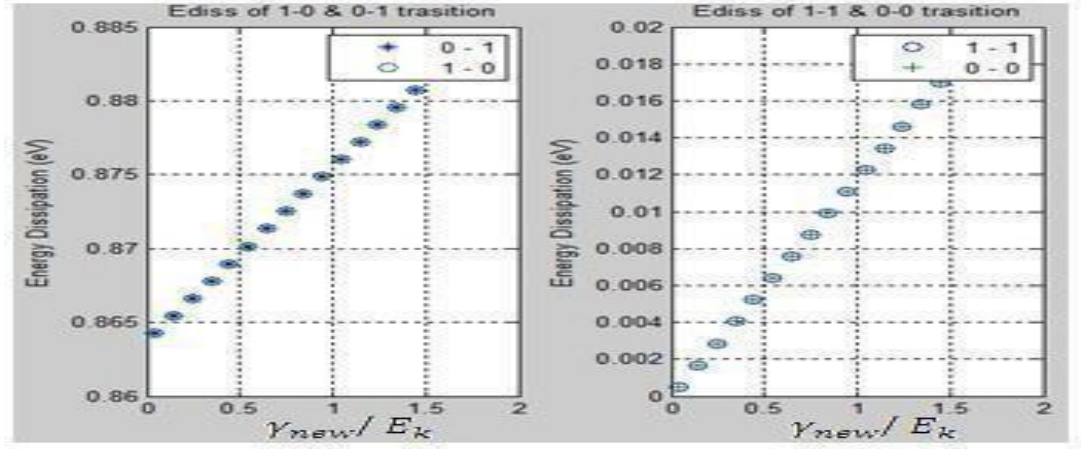


Figure 2.7: Energy Dissipation for the transition of $0 \rightarrow 0$, $0 \rightarrow 1$, $1 \rightarrow 0$ & $1 \rightarrow 1$

- If the condition is that the clock energy during switching event
- & are different then will be slightly less than but more than the kink energy, which comes from the calculation as shown in equation 4.

$$\frac{2\gamma_{new}}{E_k} \left(\frac{p_0}{\bar{p}_{old}} \gamma_{old} - \frac{p_n}{\bar{p}_{new}} \gamma_{new} \right) \dots \dots \dots (4)$$

The minimum energy requirement to dissipate the energy between the two neighboring cells is kink energy and is less than from the equation 3 because if more energy is given at the input cell, then there will be more tunneling of quantum dots from one potential well to another potential well then there is a possibility of extra energy to dissipate due to the rotational tunneling and remains same polarization as before. This rotational tunneling may not happen in the previous cells because these cells are already adjusted accordingly and it is difficult to modify them again within the QCA design. We cannot predict the electron tunneling as well as we cannot distinguish that the same electron will remain same position. In a cell of two electrons, they may exchange their position and it will happen when the applied dissipated energy is more

than the required. In that case to stabilize the energy within the cell, the electrons may rotate their position. Therefore, the energy dissipation may vary even if the state of a cell does not change (Transition from $1 \rightarrow 1, 0 \rightarrow 0$)

Chapter 3

Implementation of design of flip flop Using Quantum dot Cellular Automata

The D (Data or Delay) flip-flop is a fundamental building block in digital electronics, used for storing a single bit of data. It is known for its simplicity and reliability, making it a staple in memory elements, registers, and data storage applications. The D flip-flop ensures that the output state matches the input state at the moment of the clock edge, effectively "capturing" the input data.

3.1 D flip flop

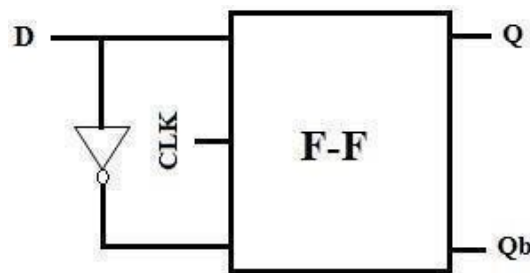


Fig 3.7: block diagram of D flip-flop majority gate structure

Inputs:

Data (D): The input data to be stored.

Clock (CLK): The clock signal that controls when the input data is sampled and transferred to the output.

Outputs:

Q: The primary output, representing the stored bit.

Q' : The complementary output, always the opposite of Q .

Operation

The D flip-flop operates by sampling the input data (D) on the rising (or sometimes falling) edge of the clock signal (CLK). When the clock edge occurs:

Clock (CLK) Rising Edge:

The value at the Data (D) input is transferred to the output (Q).

The complementary output (Q') is set to the opposite of Q .

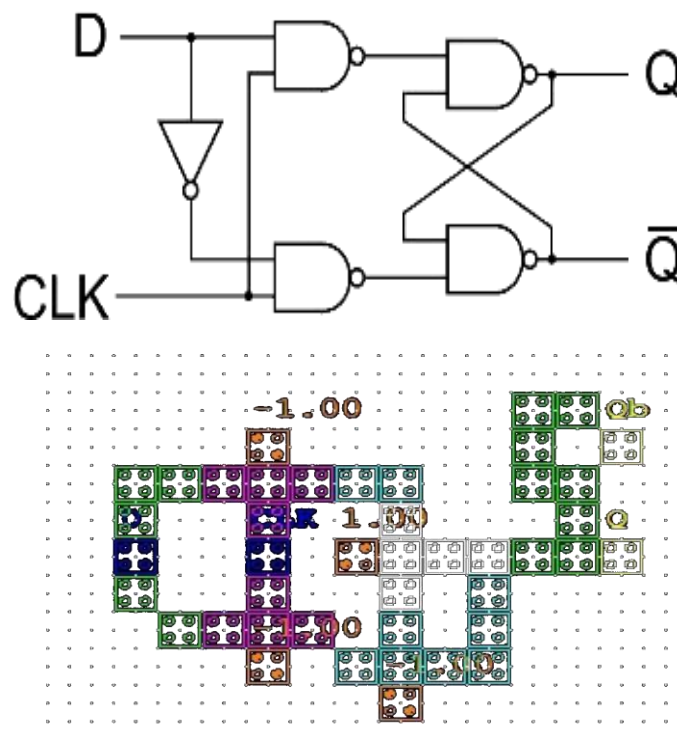


Fig 3.8 layout diagram of D Flip-Flop

The above shown diagram is the layout diagram of D Flip-Flop in Quantum-dot Cellular Automata (QCA) technology, meticulously crafted for nanoscale efficiency and precise digital

logic operations. Quantum dots are strategically arranged in a grid-like structure to mimic the T flip-flop's behavior, with dedicated regions for Toggle (T) inputs. Clocking regions synchronize state transitions, ensuring accurate data toggling. Interconnects facilitate seamless signal propagation between QCA cells, inputs, and clocking elements. Feedback loops maintain stable states, preventing unwanted transitions. The compact layout optimizes space utilization, crucial for dense circuit integration. Through thorough simulation and optimization, the T flipflop layout achieves reliable performance, low power consumption, and high-speed operation in QCA-based systems.

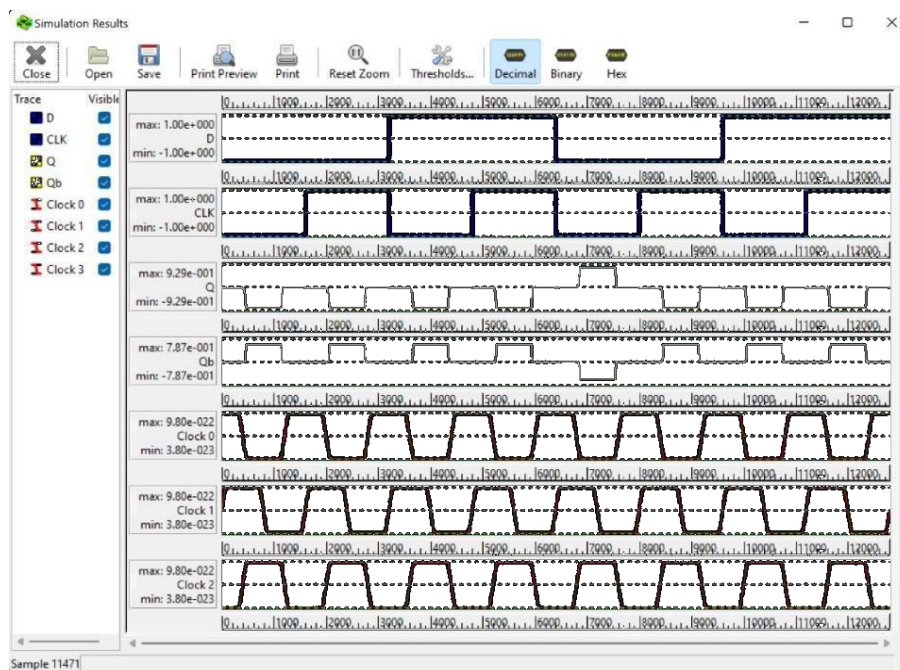


Fig 3.9 Simulation results of D flip flop

CHAPTER – 4

1. Introduction to QCA tool *Layout of QCA Basic Elements*

This section explains the layouts of basic components such as wire, 3-input majority gate, and inverter (see Fig. A.1). Figure A.1a shows cascade of QCA cells which represents a QCA wire. In a coplanar design only one layer allowed and the default layer is “main cell layer” (see the box above design area). The single cell can be created by clicking “Cell” followed by left click on the design window. Note a left click in the design area only append one cell. So we can get desired number of cells by left clicking as many times as we need. An array of cells can be obtained by clicking “Array” and dragging the mouse in the design area to a require length. To delete one or more cells click “Select” followed by selecting unwanted cell/cells and pressing delete button. Figure A.1b shows layout of a 3-input majority gate as described by $M(A, B, C) = MV3 = AB + BC + AC$. The input/output mode of a cell can be identified by double clicking corresponding cell which pop up “Cell Function” window. All the cells in a QCA circuit are clocked appropriately using “Clock Box” above design area and choosing correct clock zone from the drop list. The layout design for an inverter is obtained in similar a manner. QCA structure of an inverter is illustrated in

Fig.A

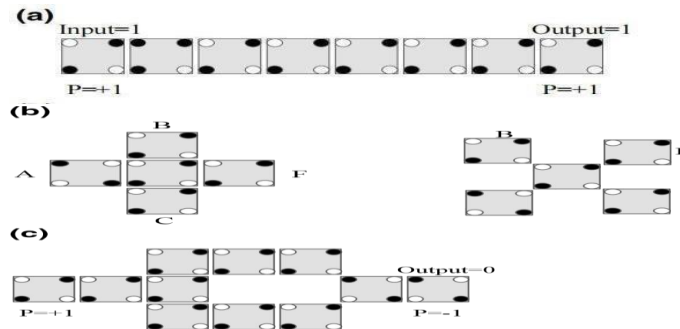


fig 4.1 QCA primitives

2. Way to Simulate:

QCA Designer 2.0.3 enables user to specify type of simulation engines: Coherence vector and bistable approximation are depicted in Fig.A.2. In addition to this, it allows user to specify the way in which the input is to be provided. To select simulation engine, click **Simulation** and select **Simulation Engine Setup**, as shown in Fig.A.2a. The default engine appears for simulation is “bistable approximation” (see Fig.A.2b). In bistable approximation engine, designer can change number of samples, maximum iteration per sample, etc. Figure A.3 shows, the coherence vector engine where designer gets the opportunity to set temperature, relaxation time, total simulation time, etc. In general, bistable approximation engine is preferably used for simulations of basic circuits, whereas coherence vector type is appropriate for studying thermal robustness of designs. To select different formats of input, click **Simulation** and select **Simulation Type Setup**, as shown in Fig. A.4a. The vector table simulation type allows designer to define the input vectors (see Fig.A.4b). User can easily insert a new vector by clicking on the “+” button. In exhaustive simulation type, considers all possible combinations of the input vectors to simulate the circuit. After the desired input type is specified, click “Start Simulation” to initiate the simulation.

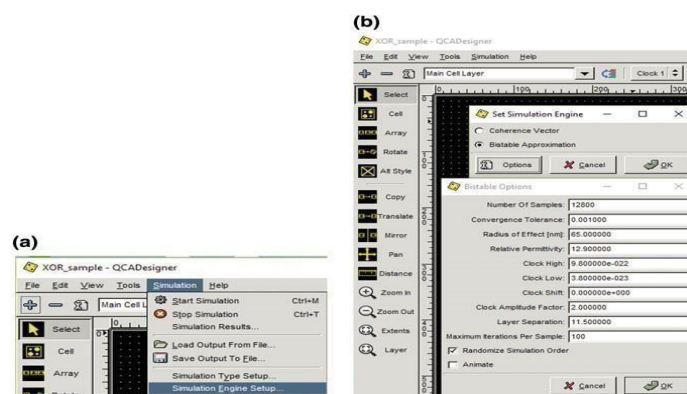


Fig . 4.2 Type of simulation engines

3. Clocking a Device

In QCA, all the cells are clocked to controls the data flow and to supply power for the cells. Different parts of the QCA layout are assigned with appropriate clock zones for correct functionality of the circuit. For illustration purpose, a 2-input XOR QCA layout is considered given by $OUT = I_0 \oplus I_1$, where I_0, I_1 are inputs (see Fig. 4.2). Both the inputs are assigned to clock zone 0. In addition, clock zone 0 also drives two inverters that yield I_0 and I_1 . Thereafter, clock zone 1 is assigned to two majority gates that realize two main terms $I_0 \cdot I_1'$ and $I_1 \cdot I_0$. Final output needs one majority gate that implement OR gate is assigned to clock zone 2. Note that one of the inputs of AND gate and R gate is indicated by polarization -1 and $+1$, respectively.

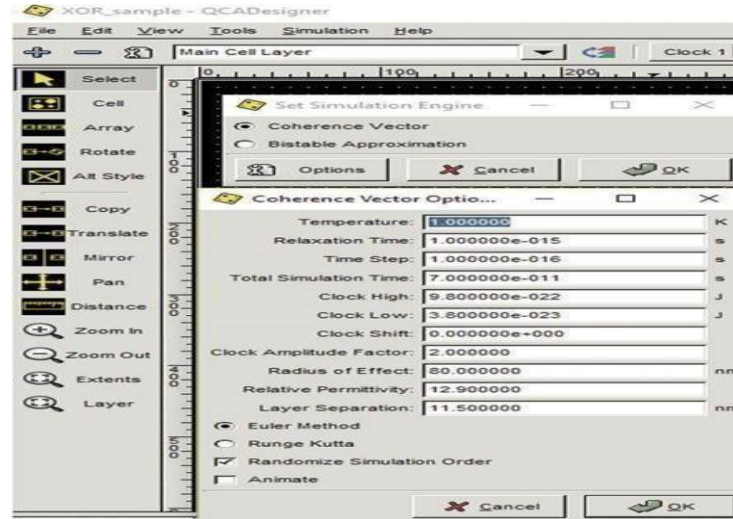


Fig. 4.3 Coherence vector engine

4. QCA Layout Evaluation Parameters

Once the layout of the circuit is completed, one can extract parameters like cell occupied area, number of cells, and delay for the corresponding design. Figure 4.6 shows selection of all the cells of the design by making a box. Thereafter, the total cell count and cell area display at the bottom of the window. Total delay of the circuit is calculated from the number of clock zones, for instance, the delay for the given circuit in Fig. 4.5 is 0.75 clocks or 3 clock phases. Note

designer can increment a clock zone of an array of cells or whole layout via “Tools → Increment cell clocks.”

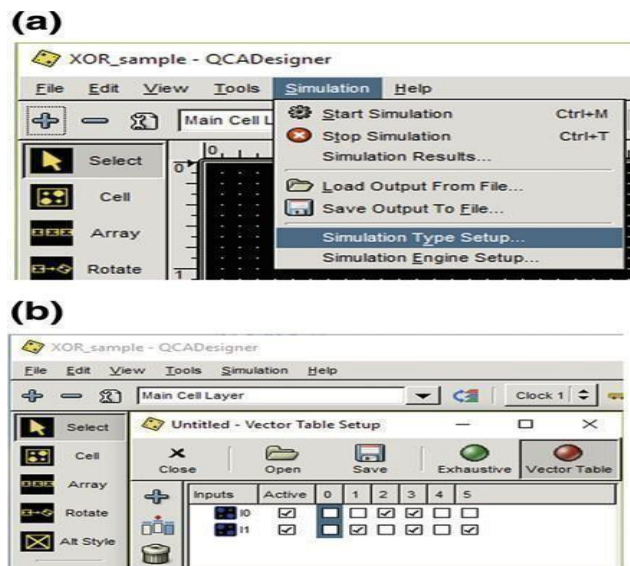


Fig. 4.4 Simulation Type Setup

5. Storing Layout and Simulation Result

To save the layouts select File → Print → Printer. The window of Fig.4.17 will be opened. In addition, print option enables setting like margins, scale, and printed objects. For coplanar layouts, deselect substrate and drawing layer in **Printed Objects**. Whereas, multilayer layouts keep all the three layers active. In the file name field of Fig.4.18, enter the desired circuit layout name. In this case, the file name used was **sample.eps** or **sample.qca** (Note: .qca file for QCA layout and .eps file for print the layout as an image file). Further, the simulation waveform can be saved by selecting “Simulation → Start simulation → Save”. While the waveform in Fig. 4.9 can be saved as image file (.eps) by selecting “Simulation → Start simulation → print

6. *Summary*

This tutorial briefly describes QCADesigner 2.0.3, a tool for QCA-based circuits. The steps provided in this section describe different crossover, how to extract design metrics, how to set up the simulation engine, and how to save the results. Please refer QCADesigner (<http://www.mina.ubc.ca/qcadesigner>) for additional documentation.

Chapter-5

Implementation of SIPO & PIPO SHIFT REGISTERS

Introduction:

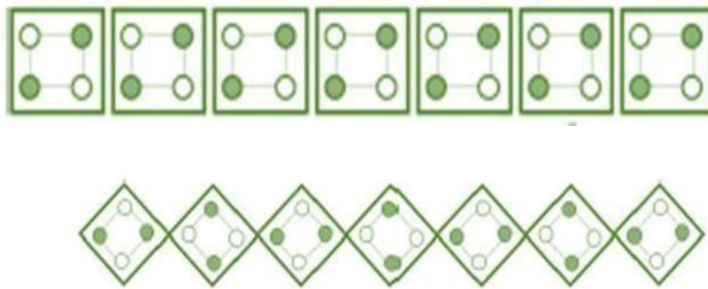
CMOS technology may be severely limited by the small geometry device characteristics, which can severely restrict transistor operation. Sub-threshold conduction, DIBL, punch-through effect, and hot carrier channel effect are some of the small geometry conditions that affect the transistor performance and can damage the device permanently. Interconnect damage caused by electromigration, electrostatic discharge, and electrical overstrain are also considerable apprehensions for the reliability of small geometry devices [1,2]. Researchers are moving towards substitute technologies and QCA nanotechnology is one of the promising technologies in this area [3,4]. The fundamental principle of the exchange of information or the information flow in QCA technology is quantum mechanical tunneling [5–9]. There is a repulsive effect between the electrons and is of among them. These electrons try to occupy the antipodal sites due to the mutual columbic repulsion. A cell in QCA is defined by its polarization state, and two polarization states exist that define either ‘0’ logic or ‘1’ logic [17–20]. A basic cell in QCA and how it occupies its two polarization states is depicted in Figure

1. This cell in QCA helps in forming other basic structures like a basic QCA wire. The first cell acts like a driver cell which affects the polarization of the cell next to it and so on. Cells adjust electrons so that there is minimum columbic repulsion between them, and information is transmitted down the QCA wire in this way [21–24]. Figure 2a,b show the two types of QCA wires i.e., the 90° wire and the 45° wire, respectively. From a 45° wire, the complement and uncomplimentary values of a signal can both be obtained. A majority voter gate and an inverter

gate are two important components in the design of the circuits in QCA. These are the basic blocks for building circuits in QCA. A majority voter gate with three inputs in QCA is presented in Figure 3 [23,25,26].



Figure 5.1. Basic QCA cell and its polarization states [27].



In QCA, clocking plays a substantial part in the proper working of the circuits. It is responsible for the control of data flow in a circuit. In addition to this, it acts as a supply and delivers power to the cells [6,10,14]. The clock is applied in four zones to a circuit in QCA. Each of these zones is represented by a different color. Green is for clock 0, clock 1 is represented by magenta, clock 2 by blue, and clock 3 is represented by white. Each clock zone has four distinct phases, i.e., switch, hold, release, and relax. Figure 4 depicts these different clocking zones in

2. Literature Review

A flip flop is a sequential circuit that has output dependency on its present input and the past output. It is a memory unit that has the capability of storing 1 bit of binary information [6]. For the realization of more complex sequential circuits, flip flops and memory cells act as the characteristic building blocks. This section presents the various D flip flop designs existing in the literature. These designs are discussed in [28–31]. The drawbacks of these designs were

highlighted, and a performance comparison was drawn based on the various performance parameters of QCA. The design discussed in [28] used 59 cells with an area of $0.075 \mu\text{m}^2$. It showed a delay of 1.75. The design proposed in [29] used 56 cells and had latency of 2.5. Hence the cell count, as well as the latency for this design, were large along with the utilization of more area, which increased overall quantum cost. The D flip flop design proposed in [30] utilized 48 QCA cells but showed a lower latency equal to 0.75. Their design is shown in Figure 5. Further, a four-bit PIPO shift register was designed using this D flip flop design that consisted of 260 cells in its QCA design and showed a latency equal to 1

Proposed Shift Register Designs

Flip flops and registers form the essential constituent. Many of the digital circuits include special purpose processors and the design of memory chips. DFF and shift registers can further be used for designing counters. In this work, new designs for different types of registers using the proposed DFF were proposed.

5.1. PIPO Designs

When the inputs are applied in a parallel manner to the register and outputs are obtained in a parallel manner, the register is called a parallel-input-parallel-output shift register. The block diagram of a simple two-bit PIPO formed using three D flip flops is shown in Figure 10. It can be seen that all inputs enter the respective flip flops in a parallel manner and that all the outputs are obtained parallel too.

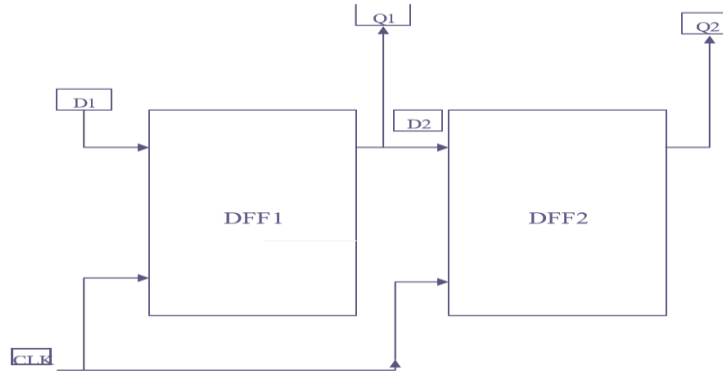


Figure 5.2. Block diagram presentation of 2-bit PIPO shift register

The two-bit PIPO shift register was constructed using two proposed DFFs. The proposed DFF design was used here. It utilized 55 cells in its QCA design. The two parallel inputs were *D1* and *D2* for stage 1 and stage 2, respectively. *CLK* is the clock signal applied to the circuit. *Q1* and *Q2* were two parallel outputs from stages 1 and 2, respectively. Each stage had a latency of 1 because the QCA clock zone changes four phases from respective input to output with a delay of 0.25 for each zone. Figure 11 shows the two-bit PIPO shift register design as employed in QCA, and Figure 12 shows its simulation graph. It can be seen that when the positive level of the clock appears, the outputs start to follow their respective inputs with a delay of 1 for each stage. For the first clock, the input *D1* is 0011, and it can be seen that output *Q1* reflects the same sequence 01 with the delay. The second input *D2* is 0101 for this duration, and output *Q2* reflects the same sequence with the delay of 1.

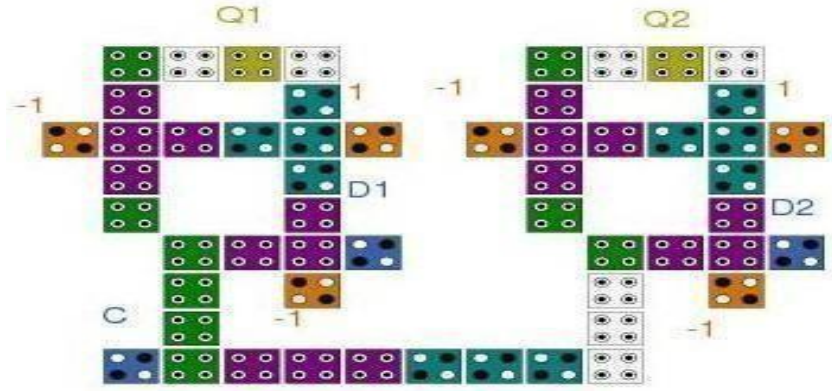


Figure 5.3. QCA implementation of 2-bit PIPO shift register.

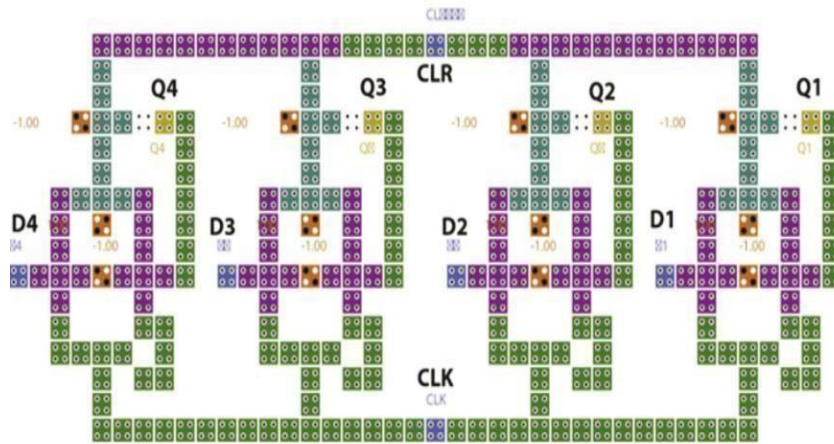


Figure 5.4. 4-bit PIPO shift register from [30].

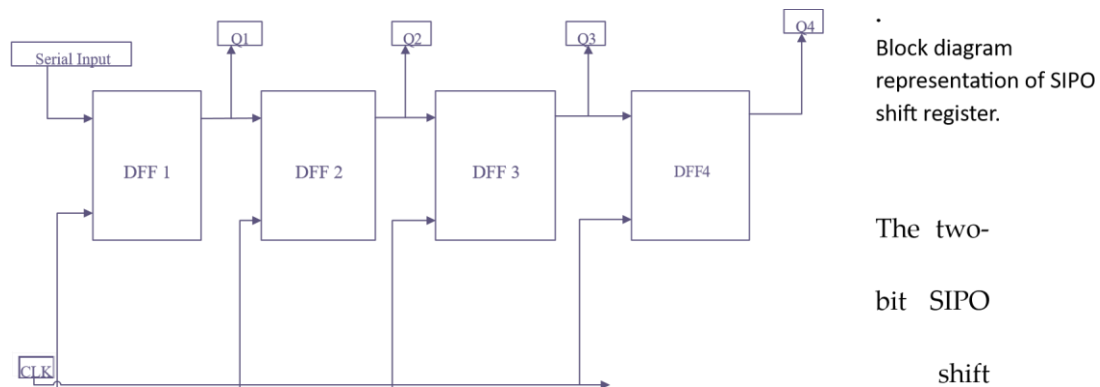
The design proposed in [31] used 44 cells and showed the latency of 1. The design discussed in [13] used 54 cells in its QCA design and increased delay equal to 1.25. This used a larger cell area and a larger total area along with the increase in total quantum cost. The design proposed in [19] utilized 24 cells in its QCA design. It occupied a cell area of about $0.007776 \mu\text{m}^2$ and showed latency equal to four clock phases, i.e., one clock cycle. Further, a four-bit shift register was designed using the presented flip flop design. It utilized 136 cells in QCA design utilizing a total area of about $0.454 \mu\text{m}^2$. The latency for this design was 1, and the quantum cost was 0.454. From the broad literature review of on flip flops designed in QCA and the realization of other more complex sequential circuits using those designs, it is found

that there is a prerequisite for optimized designs of the shift register and other sequential circuits in QCA nanotechnology.

4.2 SIPO Designs

When the input is applied serially and the output is obtained in a parallel manner, the shift register is termed as serial-input-parallel-output. Figure 15 is the block diagram representation

of a four-bit SIPO shift register. It can be seen that all the DFF are joined in a cascade manner where serial input is entered to the first DFF at the left and then the output of one connects to the input of the DFF following it. Since the same *CLK* pulse is applied to each of the DFF, the design is synchronous, and the output of each DFF is occupied in a parallel manner.



register is designed with the proposed design of DFF. It utilizes 73 cells in its QCA design. *D* is the serial input applied to the flip flop of the first stage. *Q1* is the first stage output, which connects to the input of the second stage of the SIPO shift register. *Q2* is the stage 2 output. The latency for stage 1 is 1 as the QCA clock zone undergoes four changes from input *D* to output *Q1*. These changes are clock 1 shown in magenta to clock 2 shown in blue and to clock 3 shown in white. The clock zone undergoes 4 changes from *Q1* to output *Q2*, making a total of 8 changes in clock zones from serial input *D* to output *Q2* with each change giving rise to a delay of 0.25. Therefore, the delay for the second stage is 2. Figure 16 is the implementation

of the two-bit SIPO shift register in QCA, and Figure 17 shows the simulation graph for this shift register. When the first clock pulse appears, D is 0; therefore, $Q1$ follows the serial input and remains 0 until the next clock pulse appears. It can be seen from the simulation graph that output appears after a delay of 1 in $Q1$. Then, at the second clock pulse, $Q2$ follows the input going to it through $Q1$, i.e., becomes 0 with the delay of 2, and at the same time $Q1$ goes from 0 to 1 because serial input D is 1 and has the same value until the next clock pulse appears. This process continues, and the input sequence of 0101 appears at the parallel outputs as it keeps shifting with each clock pulse.

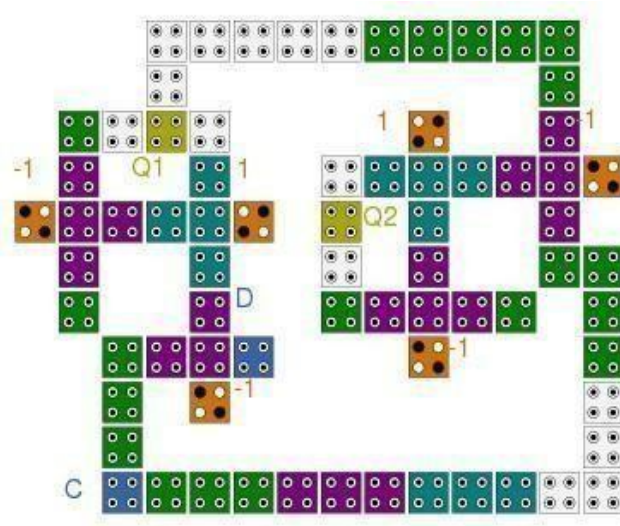


Fig 5.5 QCA Implementation of 2-bit SIPO shift register

Figure 20 shows the QCA implementation of four-bit SIPO shift register. Four proposed DFF are used to design this four-bit SIPO shift register. It utilizes 199 cells in its QCA design. D is the serial input that connects to the input of DFF in the first stage. $Q1$, $Q2$, $Q3$, and $Q4$ are the parallel outputs of four stages. Output $Q1$ of the first stage has a connection to the input of the second stage, output $Q2$ has a connection made with the input of the third stage, and output $Q3$ has a connection to the input of the fourth stage. The delay is 1, 2, 4, and 6 for stages 1, 2, 3, and 4, respectively. Figure 21 shows the simulation graph of the four-bit SIPO shift register. It

can be seen from the waveform that when the first clock pulse appears, serial input D is 0; therefore, output $Q1$ follows the input and attains the value of 0 after the delay of 1. It remains at the same value until the next clock pulse appears, and then input D is 1; therefore, $Q1$ goes from 0 to 1. At the second clock pulse, the first bit, 0, is shifted to stage 2 and is shown in the graph after the delay of 2. It remains at the same value until the third clock pulse appears and then follows the next value of $Q1$, i.e., 1. Similarly, bit 0 shifts to stages 3 and 4 at clock pulses 3 and 4. While the first bits keep shifting from stage 1 to 4, $Q1$ goes from 1 to 0 and then 1 at every clock pulse as the serial input D appears in the same manner.

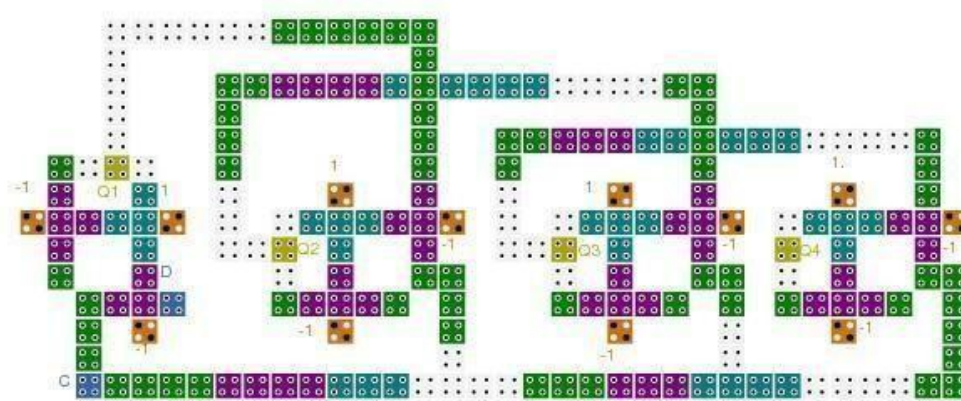


Figure 5.6. QCA Implementation of 4-bit SIPO shift register

Chapter-6

Results And Discussions

D-Flip-Flop

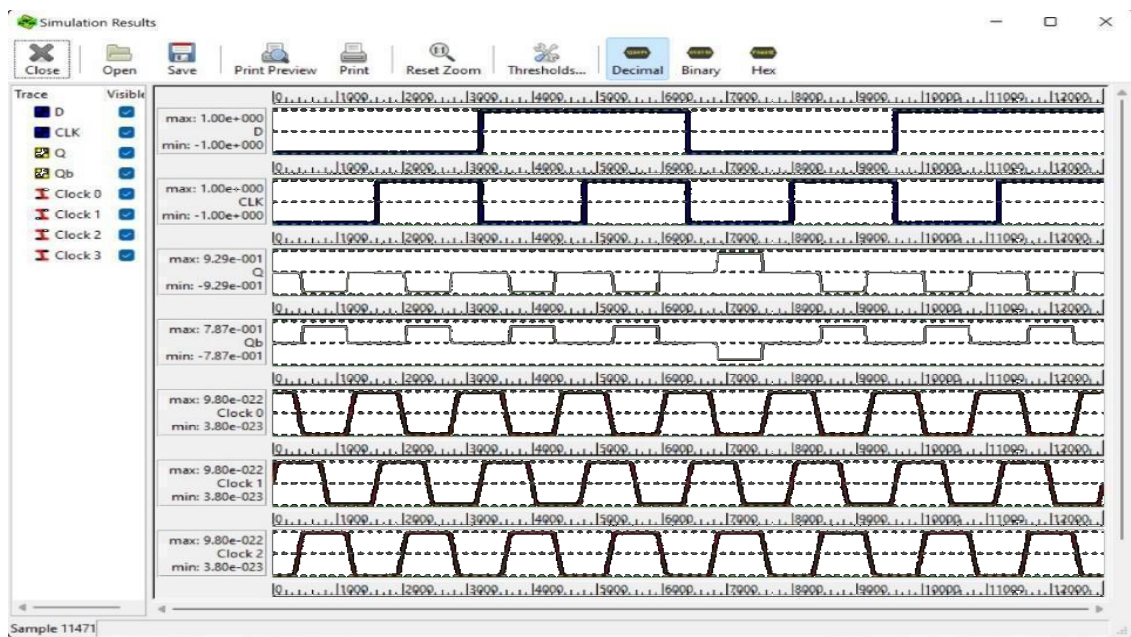


Fig 6.1 Simulation results of D flip flop

The simulation of the D (Data) flip-flop in Quantum-dot Cellular Automata (QCA) technology showcases its exceptional performance and distinct advantages over traditional CMOS technology. The input and output results highlight the efficiency and reliability of QCA-based digital circuits.

Input and Output Results:

Data (D) Input: The state of the Data input is directly transferred to the output (Q) at each clock cycle, ensuring that the output matches the input.

Clock Signal: The output (Q) updates synchronously with the clock signal, guaranteeing precise and predictable data storage and retrieval.

Hold Condition: When the clock signal is not active, the output (Q) retains its current state, demonstrating the flip-flop's ability to store data reliably.

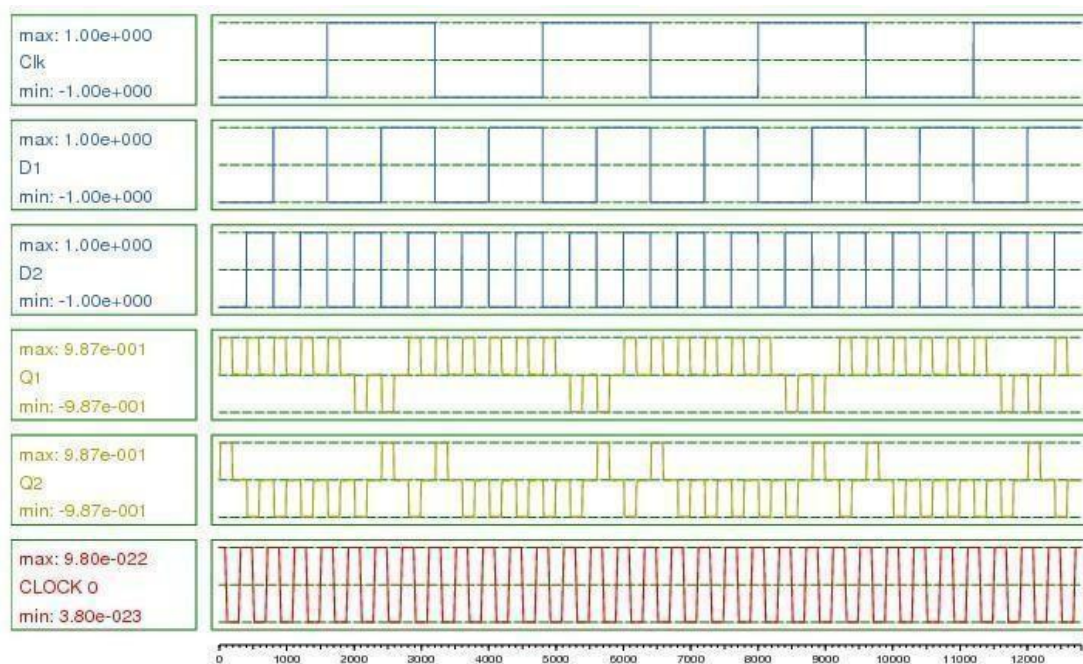


Figure 6.2 Simulation graph of 2-bit PIPO shift register.

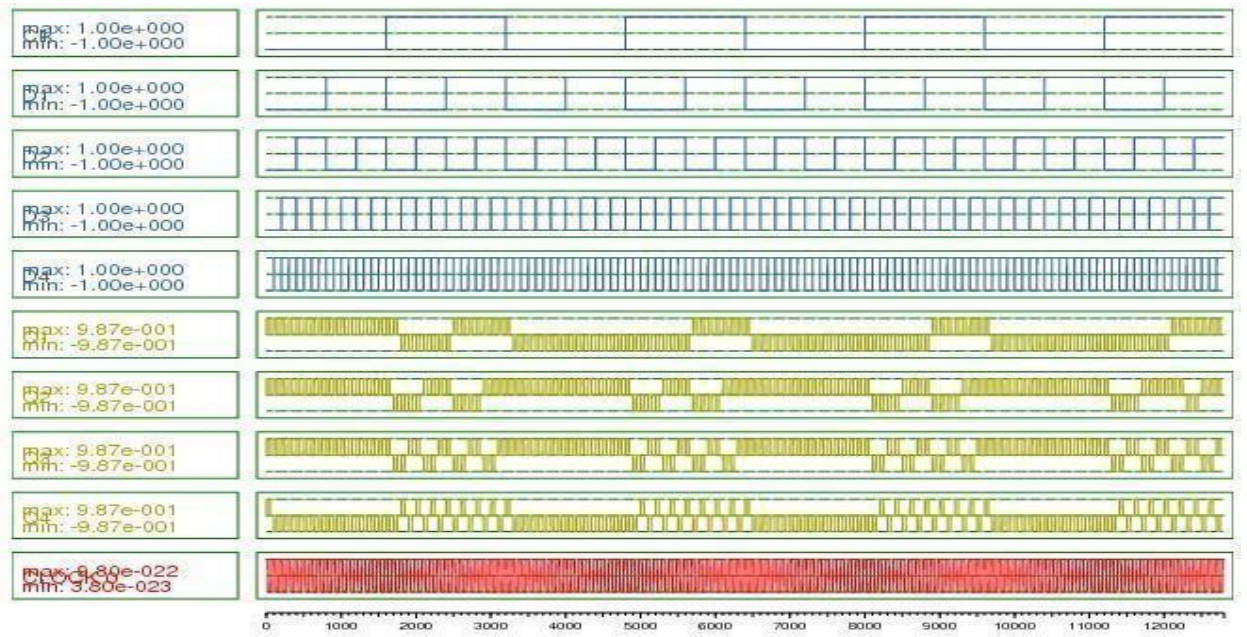


Figure 6.3 Simulation graph of 4-bit PIPO shift register.

The performance parameters of the proposed two-bit PIPO and four-bit PIPO shift registers are given in Table 1. The PIPO designs are effectual in terms of fewer total cells utilized in the design, the area occupied by the cells, total area, cost, and latency.

Table 1. Performance parameters of proposed PIPO shift registers.

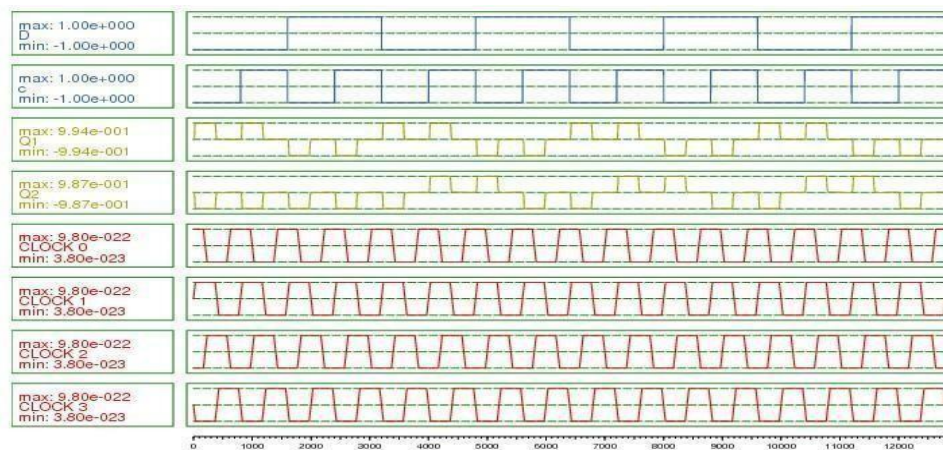


Figure 6.4 Simulation Graph of the 2-bit SIPO shift register.

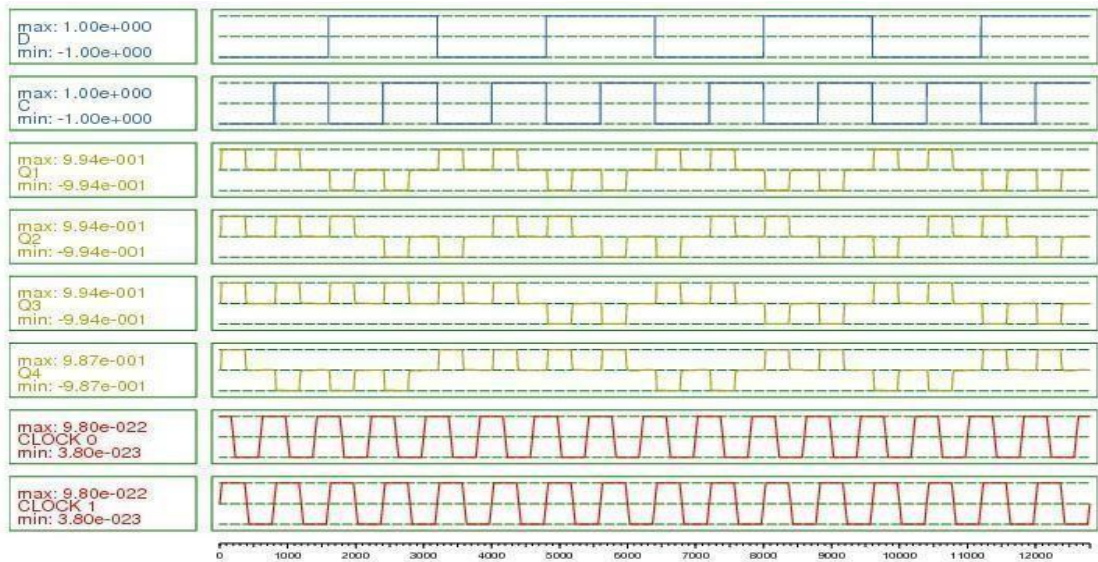


Figure 6.5 Simulation graph of 4-bit SIPO shift register.

The performance parameters for the anticipated 2-bit, 2-bit, and 4-bit SIPO shift registers are given in Table 2. These SIPO shift registers are quite efficient in terms of various QCA performance constraints such as number of QCA cells employed in the design, cell area utilization, total area utilized, and cost.

C	D	E	F	G	H	I	J	K	L
Input	Output	Total Cells	x axis distance	y axis distance	total cell area	total pan area	Total Energy Dissipation (eV)	Average Energy Dissipation (eV)	Error(+/-)
2	2	73	280 nm	220 nm	23,652 nm ²	61,600 nm ²	2.41E-02	2.19E-03	-6.87E-04
2	4	198	680 nm	280 nm	64,152 nm ²	190,400 nm ²			
3	2	55	260 nm	180nm	17,820 nm ²	46,800 nm ²	1.90E-02	1.72E-03	-5.99E-04
5	4	144	530 nm	180 nm	46,656 nm ²	95,400 nm ²			
				Cell Density=324 nm ²					

Figure 6.6 Factors comparison

Advantages Over CMOS Technology:

Lower Power Consumption: QCA technology operates with significantly reduced power consumption compared to CMOS, making it highly suitable for energy-efficient applications.

Higher Speed: The rapid state transitions in QCA-based D flip-flops lead to faster operational speeds, enhancing the overall performance of digital systems.

Compact Size: QCA allows for the development of extremely small and densely packed circuits, addressing the scaling limitations faced by CMOS technology.

Enhanced Performance: The simulation results indicate that QCA D flip-flops exhibit lower signal propagation delay and greater stability, resulting in superior overall performance.

In summary, the simulation results of the D flip-flop in QCA technology demonstrate its significant advantages over CMOS technology, including lower power consumption, higher speed, compact size, and enhanced performance. These benefits highlight the potential of QCA technology to revolutionize digital electronics in the future.

Chapter-7

Conclusion and future scope

In this project we have developed a standard equation from SR flip flop and using that equation other flip flops like D, T and JK flip-flops are subsequently designed. This is a new approach of designing flip-flops with less hardware complexity in nanotechnology. Any memory storage device can be built using the flip-flops designed in the above mentioned approach. The layout has been generated and simulation results are verified using QCA Designer simulation tool. The stability of the circuit has been clearly determined by the 3-D plots of kink energy of the two possible combinations of the output cell. In future there is a scope for design other sequential circuits like registers, counters, memory blocks and other flip flops using this generalized block.

In the future, design can be streamlined by reducing the number of cells required through approximate techniques making QCA technology more appealing compared to CMOS .QCA offers superior area, delay,and efficiency over cmos technology, suggesting that in the future, cell count can be further minimized in QCA technology.

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